

```

import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

# -----

# Load Iris dataset

# -----

iris = sns.load_dataset('iris')

# -----

# 1. Dataset overview + feature types

# -----

print("First 5 rows of dataset:\n")

print(iris.head())

print("\nData Types of Features:\n")

print(iris.dtypes)

print("\nFeature Type Classification:")

print("Numeric Features: sepal_length, sepal_width, petal_length, petal_width")

print("Nominal Feature: species")

# -----

# 2. Histograms for each feature

# -----

iris.hist(figsize=(10, 8))

plt.suptitle("Histograms of Iris Dataset Features")

plt.show()

# -----

# 3. Boxplot for all features

```

```
# -----
```

```
plt.figure(figsize=(10, 6))
```

```
sns.boxplot(data=iris)
```

```
plt.title("Boxplot of Iris Features")
```

```
plt.show()
```

```
# -----
```

```
# 4. Boxplots by Species
```

```
# -----
```

```
features = iris.columns[:-1] # excluding 'species'
```

```
for col in features:
```

```
    plt.figure(figsize=(6, 4))
```

```
    sns.boxplot(x='species', y=col, data=iris)
```

```
    plt.title(f"{col} Distribution by Species")
```

```
    plt.show()
```

```
# -----
```

```
# 5. Inference / Observations
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# -----
```

```
print("\nINFERENCES:")
```

```
print("1. Dataset has 4 numeric features and 1 categorical feature (species).")
```

```
print("2. Histograms show sepal and petal measurements are fairly distributed.")
```

```
print("3. Boxplots show presence of outliers mainly in sepal_width.")
```

```
print("4. Petal length and petal width clearly separate the species.")
```

```
print("5. Setosa species is clearly distinguishable from others.")
```

```
print("6. Virginica has the highest petal dimensions among all species.")
```

```
print("7. Versicolor lies between Setosa and Virginica in all measurements.")
```